Regression approach to representation

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1 Data

1.1 Loading

See `20260521.qmd` for reference. Like before, the stimulus labels are identical across all files, though the block order may differ. For this data, subject info is aggregated, and I cannot confidently identify which subject each recording site belongs to. Also, like before, I removed all `NA` sites and converted the binned MUAs to `qs2` files for faster loading.

1.2 Predictors

As review, the stimulus labels are:

- `coherence`: `0` when blank, `1` when noisy, `2` when naturalistic.
- `family`: `0` when blank, two or three digit integer code otherwise.
- `id`: `0` when blank, integer from `1` to `15` otherwise.
- `id_blank`: integer from `1` to `100` if blank, `0` otherwise.
- `repetition`: column constructed from counting the repetition of `coherence`, `family`, and `id` or `id_blank` combo.

For regression, unique values of `id`, `id_blank`, or `repetition` can largely be ignored. These can basically be considered repetitions of the unique `coherence` and `family` combinations. There are 36 unique family values, including the 35 texture families and a blank family.

Besides the above stimulus labels, the `block_order`, which takes values 1 through 8, might also be informative. If needed, indicators like `is_first` or `is_last` might be a working simplification for `block_order`.

So, the current set of predictors are:

- `region`: categorical, with values `"V1"`, `"V2"`, `"V4"`, and `"IT"`.
- `coherence`: categorical, represented as dummy variables `is_noisy` and `is_natur`, with `is_blank` implicit.
- `family`: categorical, represented as 35 dummy variables.
- `block_order`: categorical, represented as 7 dummy variables.

Within each unique region, there are 6300 texture observations and 600 blank observations. Omitting `block_order`, there are 90 repeated observations for each combination of `coherence` and `family`. The distribution of `block_order` is not uniform for these combinations.

1.3 Potential predictors

Unfortunately, according to Gerick, the parameters used in the texture generation model are probably lost. However, he mentioned that it should be possible to get the original stimulus images, i.e., the images presented during the passive fixation recordings. If I get these images, I should be able to design a few continuous predictors based off of similar statistics to the Portilla and Simoncelli parametric texture model, e.g., spatial correlations, luminance variance, etc.

2 PSTHs

PSTHs taken over all stimuli and sites are plotted in Figure 1.

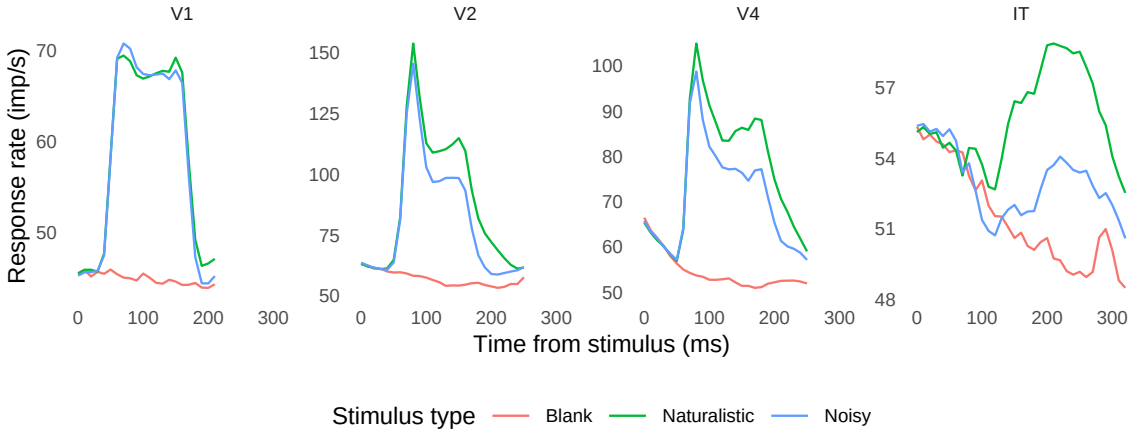

Figure 1: Averaged MUA for blank, noisy, and naturalistic textures for each region.

3 Onset times

4 Naive baseline model

4.1 Demo on single region (V1)